Collaborative Real Estate Platform – Data Pipeline 1 Project Report

1. Introduction

This project involves designing and implementing an XML-based data system for a real estate platform. It covers schema design, sample data creation, and various use-case driven transformations using XSLT. The goal is to simulate how a real estate platform would handle property listings, bookings, transactions, and services in a structured and reusable way. Outputs are generated in HTML, XML, and JSON formats.

2. Group Details & Contributions

Student Name:

Ashish Singh (Contribution – 35%)

Durga Bhavani Kowrada (Contribution – 35%)

Ravi Chandan Kodijuttu (Contribution – 30%)

Tasks completed: XML schema design, XML dataset creation, XSLT transformations for all 7 scenarios, JSON conversion, report writing.

3. Working Environment & Tools

Editor: Visual Studio Code
Extensions: XML Language Support (Red Hat), XSLT/XPath, Live Server
Online Tools: FreeFormatter (XSLT, JSON validator), JSONLint
Languages & Technologies: XML, XSD, XSLT, XPath, JSON

4. Modeling Design

The XML schema is modular, with clearly separated complex types for each major entity (Users, Properties, Agencies, Services, Bookings, Transactions). The schema uses fine-grained types, such as xs:decimal for price, xs:date for scheduling, and xs:enumeration for controlled vocabularies (e.g., user roles, property types). Required fields are implicit by default, and optional elements are explicitly marked using minOccurs=0.

The following table summarizes the structure and key attributes of the main entities in the system:

Entity	Key Attributes	Relationships / Notes
User	UserID, Name, Email, Role	Linked to Bookings, Transactions; Role can be Buyer, Seller, Agent
Property	PropertyID, Type, Location, Price, Status, AgentID	Referenced in Bookings and Transactions
Booking	BookingID, UserID, PropertyID, Date, Time	Links a user to a property viewing
Transaction	TransactionID, BuyerID, SellerID, PropertyID, Date, Amount, Status	Involves users and a property
Service	ServiceID, Type, ProviderName, Contact, AvailableDates	Standalone services like inspections, repairs
Agency	AgencyID, AgencyName, AgentIDs (list of AgentIDs)	Contains multiple agents. AgentIDs refer back to users with Role = Agent.

5. Scenario Overview

Scenario	Purpose
1 – Property Listings	Display property listings in a simple HTML table

2 – User Bookings	Join users and properties to show scheduled visits
3 – Transactions	Show buyers, sellers, and properties involved in completed transactions
4 – Services	List service providers and their available scheduling dates
5 – Agent Properties	Group properties by real estate agent
6 – XML to XML	Transform internal XML to an external listing format
7 – XML to JSON	Transform property data into JSON for web APIs

6. Modeling Challenge & Solution

One challenge was linking multiple entities such as users and properties without deeply nesting elements. To address this, ID-based linking was used across flat structures. For example, a Booking references a UserID and PropertyID, and the XSLT uses XPath to fetch full user and property data. This approach keeps the schema clean and enables flexibility.

7. Team work organization

- XML Schema: Ashish, Durga
- XML Database: Ashish, Durga, Ravichandan
- XSL Transformation: Ashish, Durga, Ravichandan
- Report: Ashish, Durga, Ravichandan